SECTION C

Answer **ALL** questions on your question paper (**50 marks**)

S/N	Question	Answer
1	A typical sequence of synoptic weather patterns associated with the passage of a cold front over Cape Town is: (A) Cut-off low, cold front, southerly meridional flow, ridging south Atlantic high (B) Coastal low, cold front, southerly meridional flow, ridging south Atlantic high (C) Coastal low, tropical cyclone, cold front, ridging south Atlantic high (D) Tropical low, cold front, tropical temperate trough, tropical cyclone (E) Low pressure ridge, coastally trapped wave, elevated inversion	
2	Assume you are visiting a friend in Beira, on the east-facing coastline of Mozambique in the Southern Hemisphere, and a tropical cyclone approaches from the east. The strongest onshore winds will most likely occur to the ___ of Beira: (A) east (B) west (C) north (D) south	
3	Deep Convection: (A) Is most prevalent at the poles (B) Occurs exclusively during the day (C) Transports heat and moisture from the surface to the upper troposphere (E) Primarily transports heat downwards to the Earth's surface	
4	The ice-albedo feedback is an example of a: (A) Radiative equilibrium model (B) Negative feedback loop in the Earth's climate system (C) Positive feedback loop in the Earth's climate system (D) Frictional force	
5	In the Southern Hemisphere, tropical cyclones and middle latitude cyclones both: (A) have surface weather fronts. (B) have winds rotating clockwise around their centre (C) will generally move from east to west (D) are shallow low-pressure systems (E) are usually barotropic weather systems	

6	When a cyclostrophic wind occurs: (A) There is no net force acting on rotating air parcels (B) The pressure gradient force equals the centripetal force in magnitude and direction (C) The pressure gradient force equals the centrifugal force in magnitude and direction (D) Gravity is balanced by the vertical pressure gradient (E) The Coriolis force is the dominant driver of wind direction	
7	What prevents wind from following the direction of the horizontal pressure gradient force: (A) Friction with the ground. (B) Gravity (C) Centrifugal acceleration (D) The Coriolis effect.	
8	Which of the following is not correct about thermal wind? (A) Thermal wind is parallel to geostrophic wind in a barotropic atmosphere (B) Westerlies weaken with height in the mid-latitudes as thermal winds oppose them. (C) Easterly winds weaken with height at the poles as they are opposed by thermal wind. (D) Cold cored cyclones intensify with height and are deep	
9	An atmospheric layer where the relative humidity is 100% and the environmental lapse rate is between the dry and saturated adiabatic lapse rates is: (A) Stable (B) Neutrally stable (C) Subsaturated (D) Unstable	
10	Which of the following variables affects saturated mixing ratio: (A) Observed vapour pressure (B)Dew point temperature (C) Temperature (D) Geostrophic wind direction (E) The Coriolis force	
11	Under which of the following conditions would density be the highest? (A) Low altitude, high temperature, and high humidity. (B) High altitude, high temperature, and high humidity. (C) High altitude, high temperature, and low humidity. (D) Low altitude, low temperature, and low humidity	

12	If an air parcel is given a small push upward and it remains where you pushed it to, the atmosphere is said to be: (A) stable. (B) unstable. (C) in geostrophic balance (D) neutrally stable. (E) saturated	
13	Which of these weather elements ***always*** decreases with height in the troposphere? (A) Wind direction (B) Temperature (C) Air pressure (D) Relative humidity	
14	The geostrophic approximation closely approximates real winds: (A) In the atmospheric boundary layer, where friction is high (B) At the Pole, where the Coriolis deflection is maximum (C) In the upper troposphere, where friction is low (D) Over cities (E) Over oceans	
15	The average winds aloft in the mid-latitudes would change from westerly to easterly if: (A) The Earth rotated in the opposite direction (B) The Earth's rotation rate decreased (C) The Earth's albedo was equal to 0 (D) The global mean temperature increased (E) Earth was perfectly spherical	
16	If they were surrounded by the same fixed air mass, which of the following air parcels would tend to be least stable? (A) Temperature = 31°C, Relative humidity = 25% (B) Temperature = 26°C, Relative humidity = 45% (C) Temperature = 15°C, Relative humidity = 50% (D) Temperature = 32°C, Relative humidity = 55% (E) Temperature = 14°C, Relative humidity = 26%	

17	Under which circumstances could relative humidity exceed 100% without producing fog or mist: (A) The dew point temperature is lower than the air temperature (B) The air is perfectly dry (C) There are too few condensation nuclei present (D) The wet bulb temperature is too high	
18	Which of the following is considered an adiabatic process? (A) Conduction heats air immediately above the Earth's surface (B) An air parcel rising, expanding and cooling without mixing with its environment (C) Rain evaporates while falling from the clouds before it reaches the ground (D) Snow cover is sublimated into a dry air parcel moving over it	
19	Which of the following is a statement of a valid atmospheric equation of state? (A) Pressure is proportional to density times virtual temperature (B) Density is proportional to pressure times temperature (C) Virtual Temperature is proportional to density times pressure (D) Virtual temperature is proportional to mixing ratio times temperature	
20	If the near-surface air temperature equals 3°C and the dew point temperature 1°C, the lifting condensation level height is approximately: (A) At the surface (B) At the tropopause height (C) 20m (D) 200m (E) 2000m	
21	In an atmospheric column that is conditionally unstable in the lower 8km of the troposphere, with a lifting condensation level height of 3000m, for which of the following would the pressure value be lowest: (A) The lifting condensation level (B) The level of free convection (C) The equilibrium level (D) The boundary layer capping inversion	
22	Which of the following atmospheric layers will always be stable: (A) An inversion layer (B) A layer with decreasing environmental potential temperature with height (C) A layer with decreasing environmental equivalent potential temperature with height (D) The daytime atmospheric boundary layer	

23	Which of the following clouds will in general be associated with the warmest cloud tops: (A) Cumulonimbus (B) cirrocumulus (C) cirrostratus (D) marine stratocumulus (E) cirrus	
24	Where are most of the world's hot deserts situated relative to the tri-cellular arrangement? (A) At the inter-tropical convergence zone (B) Around the Horse Latitudes (C) At the centre of the Ferrel Cells (D) At the equatorward edge of the Polar Cells	
25	If the wind direction is 180° the wind is coming from the: (A)West (B) South (C) North (D) South-West	
26	The centrifugal pseudo-force is most relevant for motion that: (A) Is close to the Earth's surface (B) Continues over distances of more than 10km (C) Circulates slowly around a large anticyclone (D) Changes direction rapidly where there are tightly packed, curved isobars	
27	Which of the following types of idealised wind accounts for the Coriolis effect: (A) Sea-breeze (B) Valley breeze (C) Gradient wind (D) Cyclostrophic wind	
28	Radiation fog is most common on: (A) Autumn mornings (B) Winter afternoons (C) Summer evenings (D) Spring nights	
29	In which of the following cases would the Coriolis deflection be greatest in magnitude? (A) 10 m/s wind at the Equator (B) 5 m/s wind at the South Pole (C) 10 m/s wind at the North Pole (D) 20 m/s wind at the Equator	

30	A segment of a surface cold front extends along an essentially straight line from 30°S, 10°E to 35°S, 15°E. The wind at 950hPa at the front is westerly. The wind direction will most likely ___ with height above the surface front. (A) Veer (B) Back (C) Remain fixed (D) Become undefined, as wind speed = 0	
31	For a circular synoptic-scale high pressure cell with constant size and depth, of radius 10^4 km, with depth of 300m in the mid-latitudes at 500hPa, lasting about a week, one would expect the resulting circulation to be in approximate: (A) geostrophic balance (B) anti-cyclonic gradient wind balance (C) cyclonic gradient wind balance (D)cyclostrophic balance	
32	In relation to a short-wave Rossby wave feature in the mid-latitude upper troposphere, a surface mid-latitude cyclone is most likely to occur: (A) Along the trough axis (B) Along the ridge axis (C) Ahead (eastward) of the trough axis (D) Ahead (eastward) of the ridge axis	
33	Which of the following is an example of a thermally indirect circulation cell: (A) Hadley Cell (B) Southern Hemisphere Polar Cell (C) Northern Hemisphere Polar Cell (D) Sea-Breeze circulation (E) Southern Hemisphere Ferrel Cell	
34	Cyclones in the Northern Hemisphere are generally characterised by: (A) positive relative vorticity and negative horizontal divergence (B) negative relative vorticity and negative horizontal divergence (C) positive relative vorticity and positive horizontal divergence (D) negative relative vorticity and positive horizontal divergence	
35	~~Choose the most correct statement:~~ (A) During the night time thermal turbulence dominates and pollutants are rapidly dispersed (B) During the day time thermal turbulence dominates and pollutants are mainly spread out horizontally closer to the ground (C) During the night time thermal turbulence dominates and pollutants are mainly spread out horizontally closer to the ground (D) During the day time thermal turbulence dominates and pollutants are rapidly dispersed	

36	During the day time the atmospheric boundary layer has an entrainment zone above which: (A) entrains surface layer air into the mixed layer (B) entrains free atmosphere air into the mixed layer (C) entrains mixed layer air into the free atmosphere (D) entrains mixed layer air into the surface layer	
37	In general, during the day winds flow: (A) up-slope, down-valley and mountain to plain (B) up-slope,up-valley and plain to mountain (C) down-slope, down-valley and mountain to plain (D) up-slope, down-valley and plain to mountain	
38	Tropical cyclones gain most of their energy from: (A) Rossby waves (B) insolation on their extensive, high-level clouds covers (C) the strong winds that develop within them (D) warm ocean water and latent heat release through condensation	
39	Atmospheric circulations vary in size; which list below arranges features from smallest scale to largest? (A) Tornado, tropical cyclone, thunderstorm, Hadley cell (B) Thunderstorm , tornado, tropical cyclone, Hadley cell (C) Tornado, thunderstorm, tropical cyclone, Hadley cell (D) Tornado, thunderstorm, Hadley cell, tropical cyclone	
40	~~What is the largest anthropogenic driver of air pollution?~~ (A) Agricultural activities (B) Industrial activity (C) Domestic burning (D) Energy generation	
41	Which of the following meteorological factors tends to exacerbate near-surface air pollution the most? (A) Cut off lows (B) Large rain events (C) Temperature inversions (D) Cold winter days	

42	~~Which of the following species are anthropogenic gas-phase pollutant species? Select all that apply.~~ (i) SO_2 (ii) NaCl (iii) NH_3 (iv) CO (A) i (B) i and ii (C) i, ii, iv (D) i, iii, iv	
43	For a given pressure gradient *J* at a latitude of Φ, if the Earth completed a full rotation about its axis in 48 instead of 24hrs, the geostrophic wind speed would: (A) show no change. (B) double (C) quadruple (D) halve (E) become zero	
44	When upper-level convergence of air above a surface high pressure area is weaker than the divergence of surface air, the surface pressure will: (A) Increase (B) Stay the same (C) Decrease (D) Become negative	
45	Which of the following best describes the purpose of parameterization in climate models; parameterization is …: (A) used to obtain boundary conditions for the model (B) a tool for post-processing data (C) used to increase grid resolution (D) used for representing processes that are too small to be resolved at the model grid resolution	
46	Which of the following is a statement about weather? (A) Today is a foggy day on the Cape Flats (B) Johannesburg's average summer maximum temperatures have increased by 2K over the past century (C) Long-term winter drying is expected over the Southern Hemisphere subtropics with global warming (D) A negative phase of the Southern Annular Mode is associated with wetter summers over most of southern Africa	

47	Greenhouse gases are characterised by being efficient at absorbing _____ radiation. (A) Ultraviolet (B) Visible light (C) Infrared (D) Shorter (E) Microwave	
48	Which of the following types of models is affected by lateral boundary condition problems? (A) Global Climate Models (B) Limited Area Models/ Regional Climate Models (C) Variable Grid Models (D) Global Numerical Weather Prediction Models (E) All climate models are affected by such problems	
49	A change of state directly from a solid to a gas is called: (A) Sublimation (B) Deposition (C) Condensation (D) Evaporation	
50	On average, where does maximum heating of Earth's surface occur? (A) At the Poles (B) At the Equator (C) Over the subtropics (D) Heating is equal at all latitudes	